From Mathematical Possibility to Actuality

A Modal-Existential Argument for a Necessary Ground of Concrete Reality

A Philosophical Reconstruction

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Abstract

This article develops a conditional metaphysical argument from the distinction between mathematical possibility and concrete actuality. The central claim is not that mathematics is incomplete in a way that directly proves God, nor that a mathematical description of a universe is defective. Rather, if the relevant mathematical truths are necessary while the actuality of a particular world is contingent, necessary mathematical truths alone cannot metaphysically necessitate that world's actuality without collapsing its contingency into necessity. A further explanatory principle – a restricted Principle of Sufficient Reason (PSR) for contingent actuality – then motivates a necessary ground of actuality. The resulting view is placed in dialogue with Avicenna's Necessary Existent, Leibniz's PSR, Heidegger's question of Being, Gödel's incompleteness results, the block-universe conception of time, mathematical Platonism, modal realism, and the Mathematical Universe Hypothesis. The argument is presented as a rigorous metaphysical reconstruction, not as an unconditional mathematical proof. Its decisive vulnerabilities are made explicit: the contingency of concrete reality, the explanatory PSR, the rejection of brute actuality, and the further step from a necessary ground to the richer attributes traditionally associated with God.

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1 The Problem: Description Is Not Yet Actuality

A complete physical theory may describe a world in mathematical terms. One may even adopt the stronger thesis that a physical world *is* a mathematical structure, as proposed in different forms by structural realists and, most explicitly, by Tegmark's Mathematical Universe Hypothesis [18]. Yet a conceptual question remains:

> What, if anything, distinguishes a mathematically possible structure from a concretely actual reality?

This question should be separated from two other questions. First, it is not the question whether mathematical objects exist. Mathematical Platonism says that mathematical objects are abstract and mind-independent; on standard formulations they are also non-spatiotemporal and causally inert [11, 3]. Second, it is not merely the cosmological question of what event preceded the earliest physical event. A four-dimensional block universe can have its entire temporal history represented as one tenseless spacetime structure, in which case asking for an earlier physical event may be misplaced [4].

The question is instead ontological: why is there *concrete actuality* at all, rather than only mathematical possibility or abstract structure?

The proposal developed here is:

Core Thesis. Mathematical truths can specify or characterize possible structures. If concrete actuality is genuinely contingent, however, necessary mathematical truths alone cannot be a metaphysically sufficient, necessitating ground of that actuality. If contingent actuality has a sufficient ground, its complete ground cannot itself be merely contingent. This motivates a necessary ground of actuality.

The phrase "motivates" is deliberate. The argument is conditional on substantive metaphysical premises, especially a restricted PSR. It is therefore stronger than a loose intuition but weaker than a theorem of pure logic.

2 Four Distinctions

2.1 Mathematical existence and concrete existence

Let a mathematical structure be represented abstractly by

$$\mathcal{M} = \langle D, R_1, \ldots, R_n \rangle,$$

where D is a domain and the R_i are relations, functions, or operations satisfying specified axioms. A Platonist may say that such structures exist independently of physical instantiation. A nominalist may reject this ontology while still accepting the truth or utility of mathematics. The present argument does not require full Platonism. It requires only a distinction between:

(i) the mathematical facts that characterize a possible structure, and

(ii) the fact that some concrete reality is actual.

If that distinction is denied, one has already adopted a strong metaphysical position: something like mathematical monism, full-blooded Platonism, modal realism, or the Mathematical Universe Hypothesis. Such positions are addressed below rather than ruled out by definition.

2.2 Possibility and actuality

For a world-description W, write

$$\mathsf{Possible}(W)$$

for the proposition that W is metaphysically possible, and

$$\mathsf{Actual}(W)$$

for the proposition that W is concretely actual.

The argument begins only if these can come apart:

$$\mathsf{Possible}(W) \not\Rightarrow \mathsf{Actual}(W).$$

If every possible world is actual in the same sense, then the problem must be reformulated at the level of the total plenitude: why is the totality of worlds actual, or is "actual" merely indexical? David Lewis's modal realism is a prominent case in which possible worlds are treated as concrete worlds, though only one is "actual" relative to us [10].

2.3 Contingency and necessity

Use the standard modal operators $\Box$ (necessarily) and $\Diamond$ (possibly). A proposition p is contingent if

$$\Diamond p \wedge \Diamond \neg p.$$

Thus the proposition that a particular concrete world W is actual is contingent if

$$\Diamond \mathsf{Actual}(W) \wedge \Diamond \neg \mathsf{Actual}(W).$$

The argument does not assume that every physical law, constant, or event is contingent. It needs only the contingency of the relevant total actuality.

2.4 Description, cause, explanation, and ground

These relations should not be conflated.

- A *description* represents what something is like.
- A *cause* is usually a relation among events, substances, or states.
- An *explanation* answers an appropriate why-question.
- A *ground* is a metaphysical dependence relation: one fact obtains in virtue of another [6].

The argument concerns grounding or ultimate explanation rather than a temporal cause "before" the universe. This matters especially if spacetime is a block: the sought ground need not be temporally earlier than the block.

3 The Modal Core: Why Necessary Mathematics Does Not Select a Contingent Actuality

Let N stand for the totality of mathematical facts relevant to the structure of a possible world W. Suppose these mathematical facts are necessary:

$$\Box N. \tag{1}$$

Suppose also that the concrete actuality of W is contingent:

$$\Diamond \mathsf{Actual}(W) \wedge \Diamond \neg \mathsf{Actual}(W). \tag{2}$$

What would it mean for mathematics *alone* to be a necessitating sufficient ground of actuality? A strong formulation would require:

$$\Box(N \rightarrow \mathsf{Actual}(W)). \tag{3}$$

But from (1) and (3), standard normal modal reasoning yields

$$\Box \mathsf{Actual}(W),$$

contradicting the contingency condition in (2).

Proposition 1 (Modal Non-Selection)**.** *If the mathematical facts N are necessary and the actuality of W is contingent, then N alone cannot be a metaphysically necessitating sufficient ground of* $\mathsf{Actual}(W)$*.*

Proof. Assume $\Box N$ and $\Box(N \rightarrow \mathsf{Actual}(W))$. By closure of necessity under modus ponens in normal modal logics, $\Box \mathsf{Actual}(W)$. But if $\mathsf{Actual}(W)$ is contingent, $\Diamond \neg \mathsf{Actual}(W)$. Hence the assumptions are inconsistent with contingency. Therefore, given contingency, necessary mathematical truths alone cannot necessitate the world's actuality. □

This proposition establishes less than "mathematics cannot explain actuality" in every possible sense. It establishes a precise point: if mathematical truths are necessary and world-actuality is contingent, mathematical necessity cannot by itself *necessitate* that actuality. A non-necessitating statistical or contrastive explanation would require additional metaphysical work and would not by itself close the ultimate explanatory question.

4 From the Modal Core to a Necessary Ground

The modal core only opens an explanatory gap. To move from that gap to a necessary ground requires an additional principle.

4.1 A restricted Principle of Sufficient Reason

Leibniz's strong PSR is often summarized as the principle that no fact obtains without a sufficient reason why it is so rather than otherwise [9, 12]. Contemporary discussions distinguish stronger and weaker versions of PSR and debate whether unrestricted formulations generate modal-collapse problems [14, 13, 16].

For present purposes we use a restricted form:

Definition 1 (PSR-A)**.** Every contingent fact of concrete actuality has a sufficient metaphysical ground or complete explanation.

PSR-A does not say that every truth whatsoever has a causal explanation, nor that necessary truths are caused. It targets contingent concrete actuality.

4.2 The total contingent fact

Let C denote the total contingent fact: the conjunction or totality of all contingent facts of concrete actuality. We need not assume that C is literally a set-theoretic conjunction; the notation is shorthand for the whole contingent domain whose actuality is at issue.

By PSR-A, if C is contingent, there is some complete ground G such that

$$G \blacktriangleright C.$$

Suppose G were itself wholly contingent. Then G would belong to, or be included within, the very totality of contingent reality represented by C. If G is invoked as the complete explanation of C while being only another dependent member of C, the explanation is not yet ultimate: the actuality of G remains among the facts requiring grounding.

This point does not depend on a temporal regress. An infinite chain of contingent explanations can be temporally beginningless and still leave a synchronic or global explanatory question: why does the whole contingent structure obtain? Avicenna and Leibniz both exploit versions of this distinction, though with different metaphysical frameworks [2, 1, 17, 9].

Proposition 2 (Necessary-Ground Consequence)**.** *Assume (i) there is a total contingent actuality C, (ii) PSR-A is true, and (iii) no wholly contingent member or collection is a complete ground of the totality of contingent actuality. Then the complete ground of C includes at least one non-contingent reality.*

Proof. By (i) and (ii), C has a complete ground G. If G were wholly contingent, then by (iii) it could not be the complete ground of C. Therefore G is not wholly contingent. Hence the complete ground includes a non-contingent element or reality. In modal terms, the ultimate ground is necessary in at least the relevant ontological sense. □

The conclusion is intentionally modest:

There is a non-contingent ground of contingent actuality.

It does not yet establish omniscience, perfect goodness, personhood, or any revealed theology.

5 Avicenna: Necessary Existence and Contingent Being

Avicenna's metaphysics offers a particularly close historical analogue. In *The Metaphysics of The Healing*, he distinguishes what is necessary in itself from what is possible (contingent) in itself. A contingent existent does not contain within its essence the sufficient explanation of its existence; if actual, its existence is from another. His argument aims ultimately at the Necessary Existent, whose existence is not received from another [2, 1, 17].

Our reconstruction differs in emphasis. Instead of beginning with ordinary contingent substances, it begins with the gap between a mathematical specification and concrete actuality. But the structural movement is Avicennian:

$$\text{contingent actuality} \longrightarrow \text{ontological dependence} \longrightarrow \text{non-contingent ground}.$$

An important caution follows. Avicenna's full system contains additional theses about essence and existence, causation, simplicity, unity, intellect, and emanation. The argument developed here should not be presented as if it simply reproduces Avicenna's demonstration. It is a contemporary modal reconstruction inspired by the same contingency/necessity architecture.

6 Leibniz: Why This Reality Rather Than Another?

Leibniz sharpens the contrastive form of the question. Even if every event within an infinite world-series has an earlier explanation, one can still ask why that whole series exists and why this series obtains rather than another. His PSR is designed to prevent the explanatory chain from terminating in an unexplained contingent totality [9, 12].

In the present framework, the Leibnizian question becomes:

$$\text{Why } \mathsf{Actual}(W_1) \text{ rather than merely } \mathsf{Possible}(W_1),$$

or, if many worlds are actual,

$$\text{Why this total actuality } \mathcal{A} \text{ rather than no concrete actuality?}$$

Leibniz famously moves from possible worlds to divine understanding and will: understanding represents possibilities; will selects; power actualizes. Our argument does not build those attributes into the premises. It first seeks only the non-contingent ground. Whether selection among genuinely alternative possibilities implies something intellect-like or will-like is a further argument, treated in Section 11.7.

7 Heidegger: The Question of Being as a Constraint on the Argument

Heidegger's importance here is partly negative. The opening question of *Introduction to Metaphysics* asks why there are beings at all rather than nothing [8]. Yet Heidegger resists the tendency of classical metaphysics to treat Being as merely another entity, even the supreme entity. His critique of onto-theology warns against turning the ultimate ground into simply "the largest thing in the inventory of things."

A Heideggerian constraint on our argument can therefore be formulated as follows:

> If a necessary ground is posited, it should not be conceived as one more contingent-like object located inside the world whose actuality it grounds.

This does *not* mean that Heidegger would endorse the Avicennian or Leibnizian conclusion. He would resist important parts of their metaphysical vocabulary. The synthesis proposed here is therefore constructive and tension-filled, not exegetical:

Avicenna :	contingent being points to necessary existence,
Leibniz :	contingent actuality calls for sufficient reason,
Heidegger :	do not reduce the question of Being to another ordinary being.

The resulting proposal is to speak of a *necessary ground of actuality* before speaking of a "highest object."

8 Gödel: Incompleteness Does Not Make a Mathematical Universe Impossible

Gödel's first incompleteness theorem shows, roughly, that any consistent, effectively axiomatized formal system sufficiently expressive for arithmetic contains true statements that are not provable within that system; the second theorem limits such a system's ability to establish its own consistency under appropriate conditions [7].

These theorems do not show that a complete mathematical structure cannot exist. They distinguish a structure from a formal proof system that attempts to capture all truths about it. Thus:

$$\text{structure} \neq \text{formal theory of the structure.}$$

The natural numbers may be treated as a determinate structure even though no single suitable effective axiomatization captures every arithmetic truth, depending on one's semantics and meta-mathematical commitments. Consequently, Gödel does not refute the idea that a universe could be mathematical.

What Gödel does contribute to our discussion is a warning against a stronger thesis: that "mathematical reality" must be identical with what can be exhaustively generated by one finite mechanical proof system. Even Tegmark's discussion of a mathematical universe explicitly confronts Gödelian issues and proposes restricting attention to computable/decidable structures as one response [18].

Therefore the actuality argument should not be based on the invalid inference

$$\text{Gödel incompleteness} \Rightarrow \text{mathematical universe impossible.}$$

Instead, its independent modal claim is:

$$\text{necessary mathematical truth} \not\Rightarrow \text{contingent concrete actuality,}$$

unless one gives up the contingency assumption or adopts a plenitude thesis.

9 The Four-Dimensional Block and Timeless Actualization

In a B-theoretic or block-universe conception, all times or events can be equally real, and the fundamental temporal description is tenseless [4]. Represent a complete history schematically as

$$\mathcal{U} = \{S(t) : t \in T\},$$

where $S(t)$ denotes the total physical state associated with temporal coordinate t.

If the block is fundamental, it is misleading to imagine the universe "actualizing itself" slice by slice from an external time. The whole block is the candidate concrete object. The metaphysical question is therefore global:

$$\text{Why is } \mathcal{U} \text{ actual?}$$

not merely

$$\text{What caused } S(t+1) \text{ from } S(t)?$$

A necessary ground need not be temporally prior to the block. If timeless grounding is coherent, one may write schematically

$$G \blacktriangleright \mathcal{U}$$

without positing a time $t = -1$ at which G "switches on" spacetime.

This model also dissolves an apparent conflict between a block universe and unpredictability. A future event can be part of a complete spacetime structure even if no observer at an earlier time can compute or infer it. Ontological determination, computational accessibility, epistemic predictability, and causal determinism are distinct notions. Quantum theory intensifies these distinctions but does not by itself decide the metaphysics of time.

10 Could Every Possible Universe Be Actualized?

Suppose there is a modal space

$$\mathfrak{P} = \{W_i : W_i \text{ is genuinely possible}\}.$$

The grounding hypothesis leaves open several possibilities:

(a) exactly one world is actualized;

(b) some proper subset of possible worlds is actualized;

(c) every possible world is actualized;

(d) "actuality" is indexical, as on Lewisian modal realism, rather than a privileged metaphysical property.

If every coherent mathematical structure exists concretely, the original selection problem changes. There may be no question "Why this world rather than another?" at the global level. But a new question can remain:

$$\text{Why does the plenitude of structures have concrete existence at all?}$$

A defender of plenitudinous mathematical realism may answer that mathematical existence just *is* existence; no further actuality relation exists. That is a coherent escape from the argument, but it is not a neutral result. It is a substantive identity thesis:

$$\text{mathematical existence} = \text{concrete existence}.$$

The burden then shifts to explaining why the ordinary abstract/concrete and causal/non-causal distinctions should be abandoned or reconstructed.

11 The Strongest Objections

11.1 Objection 1: Brute actuality

The simplest objection is to reject PSR-A:

> The total concrete reality is a brute fact. It has no further explanation.

This position is logically available. Contemporary discussions of brute facts explicitly connect them to denials or restrictions of PSR [5]. Russell's famous stance in his debate with Copleston is often taken as a paradigmatic refusal to demand an explanation of the whole universe. Humean and anti-rationalist traditions likewise resist unrestricted explanatory demands.

Reply. The argument cannot derive a contradiction from brute actuality using pure logic alone. Its response must therefore be abductive or methodological: accepting brute actuality stops explanation exactly at the point where the most comprehensive contingent fact appears. A defender of PSR-A argues that this is theoretically costly, especially when we ordinarily seek explanations for contingent states precisely because they could have been otherwise. But this is a philosophical commitment, not a theorem.

11.2 Objection 2: PSR causes modal collapse

Strong forms of PSR can threaten to make every truth necessary: if the necessary ground plus sufficient explanation necessitates every actual fact, how can contingency survive? Variants of this concern have been developed in contemporary metaphysics [5, 12].

Reply. The present argument deliberately uses PSR-A rather than an unrestricted PSR. Even then, the problem is not automatically solved. If the necessary ground necessarily produces exactly W, then W appears necessary. A theistic development therefore needs a coherent account of non-necessitated grounding – traditionally, something like free creation – or an account on which the ground is necessary while its contingent effects are not. This is a genuine additional burden.

11.3 Objection 3: The universe itself is necessary

One could deny (2) and claim

$$\Box\mathsf{Actual}(W).$$

Then the modal non-selection argument does not apply.

Reply. This is coherent if defensible. But it requires showing that no genuinely alternative total reality is metaphysically possible. The mere fact that the universe has mathematical laws does not establish this necessity. The necessity thesis is therefore an alternative metaphysical endpoint rather than a refutation by itself.

11.4 Objection 4: Mathematical existence just is physical existence

Tegmark's strongest formulation identifies the physical world with an abstract mathematical structure [18]. If the identity is literal, asking what "adds actuality" may be a category mistake.

Reply. The identity thesis avoids the gap by denying the distinction on which the argument starts. But it must then explain why some mathematical structures appear as causally efficacious, spatiotemporal, observer-containing worlds while abstract mathematics is usually characterized as causally inert. One can answer by revising the abstract/concrete distinction, but that is a major metaphysical commitment, not something established by mathematics alone.

11.5 Objection 5: Modal realism

Lewisian modal realism holds that possible worlds are concrete and equally real; "actual" functions indexically, much like "here" [10]. This radically weakens the question why one possible world was selected for actuality.

Reply. It does not automatically answer why there is a plurality of concrete worlds at all. Lewis may treat the plurality as part of the best overall theory rather than as something requiring a further ground. The disagreement therefore returns to explanatory methodology and PSR.

11.6 Objection 6: A totality need not need a further explanation

The inference from "each contingent member has an explanation" to "the totality has an explanation" can commit a fallacy of composition.

Reply. The argument here does not infer the explanatory need of the whole merely from the explanatory need of each part. PSR-A is applied directly to the contingent total fact C. Whether that application is legitimate is exactly the disputed premise. Stating it explicitly prevents the fallacy from being hidden.

11.7 Objection 7: The necessary ground need not be God

Even if a necessary ground exists, why identify it with God? It could be an impersonal necessary law, field, structure, or metaphysical principle.

Reply. This objection is decisive against any premature theological conclusion. The present argument establishes, at most, a necessary ground of contingent actuality. Additional arguments are required for unity, simplicity, causal power, intellect, will, goodness, or personhood. Avicenna and Leibniz each supply extensive further reasoning; contemporary authors such as Rasmussen and

Pruss offer other routes to a necessary being [15, 14]. None of those further attributes should be smuggled into the conclusion of the present argument.

12 A Refined Argument in Twelve Steps

The entire proposal can now be stated without rhetorical shortcuts.

P1. There are mathematical truths and structures capable of characterizing possible worlds.

P2. At least the relevant pure mathematical truths are necessary rather than contingent.

P3. Concrete actuality is conceptually distinguishable from mere mathematical possibility, unless one adopts a strong identity or plenitude thesis.

P4. The actuality of the total concrete world is contingent: it could have failed to exist or could have been otherwise.

P5. If necessary mathematical truths alone metaphysically necessitated that actuality, the actuality would itself be necessary.

P6. Therefore necessary mathematical truths alone do not necessitate the contingent actuality of the world.

P7. Every contingent fact of concrete actuality has a sufficient metaphysical ground or complete explanation (PSR-A).

P8. Let C be the total contingent actuality.

P9. By P7, C has a complete ground.

P10. A ground composed wholly of contingent actuality cannot be the ultimate complete ground of the totality of contingent actuality, because its own actuality remains within the explanandum.

P11. Therefore the complete ground of C includes a non-contingent reality.

P12. Hence, conditional on P3, P4, P7, and P10, there exists a necessary ground of contingent actuality beyond merely possible mathematical description.

The inferential structure should be understood as conditional:

$$(P3 \wedge P4 \wedge P7 \wedge P10) \rightarrow \exists G\, \mathsf{NecessaryGround}(G, C).$$

The argument's strength therefore depends primarily on four questions:

(i) Is the abstract/concrete distinction genuine?

(ii) Is total concrete actuality contingent?

(iii) Is PSR-A true?

(iv) Can a wholly contingent totality be completely grounded by contingent reality alone?

These are the proper sites of philosophical disagreement.

13 What the Argument Does and Does Not Establish

It supports, conditionally	**It does not by itself prove**
A distinction between mathematical possibility and contingent actuality	That mathematical Platonism is true
That necessary mathematical truths cannot alone necessitate contingent actuality	That the universe had a temporal beginning
A necessary ground if PSR-A and the anti-circularity premise are accepted	That Gödel's theorem proves God
A timeless grounding model compatible with a block universe	That quantum mechanics is false or deterministic
A natural bridge to Avicennian and Leibnizian contingency arguments	That the necessary ground is personal, omniscient, or morally perfect
A Heideggerian warning against conceiving the ground as merely another intraworldly object	That Heidegger himself endorsed classical theism

14 Conclusion

The most defensible version of the argument is not "mathematics is incomplete, therefore God," nor "an LLM is mathematical, therefore immaterial beings exist." Those moves are too quick. The stronger route begins with a modal distinction.

If the mathematical truths characterizing a possible world are necessary, while that world's concrete actuality is contingent, mathematics alone cannot be a necessitating sufficient ground of its actuality without converting contingency into necessity. This yields a precise explanatory gap between *what can be mathematically specified* and *what is concretely actual*.

A restricted Principle of Sufficient Reason then turns that gap into a contingency argument: if the total contingent actuality has a complete ground, and no wholly contingent item or collection can completely ground the whole contingent domain, then the ultimate ground must be non-contingent. In that limited but significant sense, a necessary ground of actuality follows.

Avicenna supplies the metaphysics of necessary versus contingent existence. Leibniz supplies the demand for a sufficient reason of the total contingent order. Heidegger supplies a caution: the ultimate issue is not adequately captured by picturing a supreme object alongside other objects. Gödel supplies a different caution: mathematical structure is not identical with exhaustive derivability inside one formal system. The block-universe picture then allows the grounding relation to be timeless rather than a first event before time.

The remaining philosophical choice is sharp but unresolved. One may reject the argument by accepting brute actuality, by declaring the whole concrete reality necessary, by identifying mathematical and concrete existence, or by adopting a plenitude of concrete worlds. Or one may accept that contingent actuality requires a non-contingent ground. The argument does not force that choice by logic alone. What it does is expose exactly where the choice occurs and what each

alternative must bear.

The central question can therefore be stated in its most compact form:

If possibility does not actualize itself, what grounds the fact that anything is actual?

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